

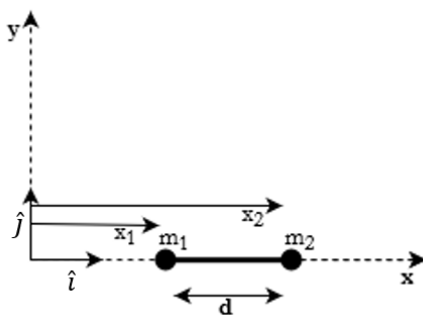
MAE 5730/4730 HW Problem #17

Two Masses Constrained

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Finding the DAE Matrix

Coordinate System:



FBDs:



LMB on m_1 :

$$\Sigma F_{m_1} = m_1 \ddot{x}_1 = F(t) + T$$

LMB on m_2 :

$$\Sigma F_{m_2} = m_2 \ddot{x}_2 = -T$$

Kinematic constraint:

$$x_2 - x_1 = l$$

$$\dot{x}_2 - \dot{x}_1 = 0$$

$$\ddot{x}_2 - \ddot{x}_1 = 0$$

DAE Equations:

1) $m_1\ddot{x}_1 + 0\ddot{x}_2 - T = 0$

2) $0\ddot{x}_1 + m_2\ddot{x}_2 + T = 0$

3) $-\ddot{x}_1 + \ddot{x}_2 + 0T = 0$

DAE Matrix:

$$\begin{bmatrix} m_1 & 0 & -1 \\ 0 & m_2 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ T \end{bmatrix} = \begin{bmatrix} F(t) \\ 0 \\ 0 \end{bmatrix}$$

where $F(t) = -f\sin(t)$

Error in Kinematic Constraint

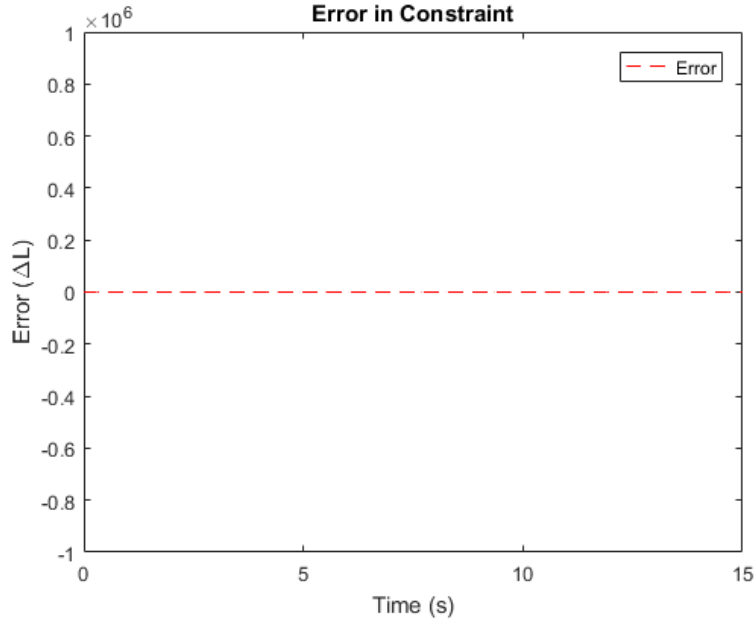


Figure 1: Accumulating Error in Kinematic Constraint

As seen in Figure 1, the constraint is satisfied as time increases, with no accumulating error.

Computation Speed Compared to Problem 16

To compare computation speed between this solution and that of Problem 16, where the two masses are attached by a spring with a high stiffness, two trials were run with identical initial conditions:

$$x_1(0) = 0$$

$$x_2(0) = 1$$

$$f = 1$$

$$k = 1000$$

To measure the time taken to find a solution, a timer is started using Matlab command `tic` before calling `ode45`, then the timer is read after `ode45` concludes using matlab command `toc`, as seen below:

```
1 tic
2 [t,z]= ode45(@TwoMassConstrainedRHS,t_span,init, options,param);
3 endTime= toc
```

The result of this trial is summarized below in Table 1

	HW16	HW17
Computation Time:	69ms	9ms

Table 1: Computation Times for 15s Simulation

Clearly, the constraint method used in Problem 17 is faster than an infinite stiffness spring as used in Problem 16. From this trial, we can estimate the constraint method $> 7x$ faster than the infinite stiffness method.

Unequal Initial Velocities

To investigate the effect of unequal initial velocities applied to the masses, the simulation was begun with the following initial conditions:

$$x_1(0) = 0$$

$$\dot{x}_1(0) = 1$$

$$x_2(0) = 1$$

$$\dot{x}_2(0) = 2$$

These initial conditions resulted in the following trajectories:

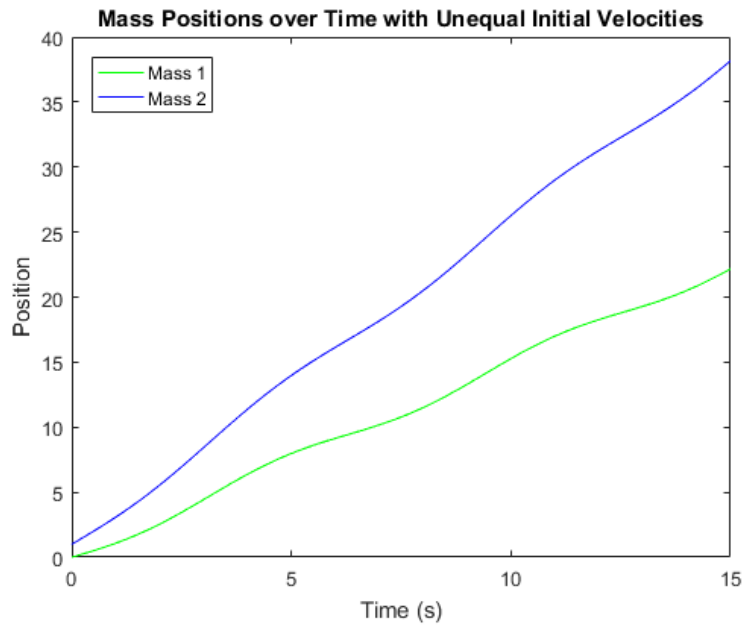


Figure 2: Mass Trajectories after Non-physical Initial Conditions

Clearly, the kinematic constraint has been broken by the non-physical initial conditions. The result of these unequal initial conditions is that both masses are acted upon as before, however they each maintain their initial velocities. Therefore, the two masses move apart from each other at a velocity equal to the difference of their initial velocities.

Matlab Code

```
1 %% MAE 5730, Problem Set 5
2 % Gabriel Rodriguez
3 % HW 17, 9/27/2017
4
5 %% Setup
6 close all
7
8 % Simulation Duration
9 t_end= 15;
10 t_span= [0 , t_end];
11
12 % Set parameters
13 m1= 1;
14 m2= 1;
15 param.A= [m1 0 1;
16           0 m2 1;
17           1 1 0];
18 param.F= 1;
19
20 % Initial conditions
21 x1_o= 0;
22 x2_o= 1;
23 inits= [x1_o;
24         x2_o;
25         0;
26         0];
27
28 l= x2_o - x1_o;
29
30 % ODE45 error tolerances
31 options= odeset('RelTol',1e-6,'AbsTol',1e-6);
32
```

```

33 %% Calculate Solution
34 tic
35 [t,z]= ode45(@TwoMassConstrainedRHS,t_span ,inits ,options ,param);
36 endTime= toc
37
38 x1= z(:,1);
39 x2= z(:,2);
40
41 dl= x2 - x1;
42 error= dl - l;
43
44 %% Plotting
45 figure()
46 plot(t,x1,'g',t,x2,'b')
47 title('Mass Positions over Time')
48 legend('Mass 1','Mass 2','Location','NorthWest');
49 xlabel('Time (s)'); ylabel('Position');
50
51 figure()
52 plot(t,error,'r')
53 title('Error in Constraint')
54 axis([0,15,1e6,1e6])
55 xlabel('Time (s)'); ylabel('Error (\Delta L)');
56 legend('Error')
57
58
59 %% Unequal Initial Mass Velocities
60 % Initial conditions
61 x1_o= 0;
62 x2_o= 1;
63 inits= [x1_o;
64         x2_o;
65         1;
66         2];

```

```
67
68 l= x2 - x1;
69
70 % Calculate Solution
71 [t,z]= ode45(@TwoMassConstrainedRHS,t_span,init, options,param);
72
73 x1= z(:,1);
74 x2= z(:,2);
75
76 % Plotting
77 figure()
78 plot(t,x1,'g',t,x2,'b')
79 title('Mass Positions over Time with Unequal Initial Velocities')
80 legend('Mass 1','Mass 2','Location','NorthWest');
81 xlabel('Time (s)'); ylabel('Position')
```